In order to correctly model the different possible states of the system, it will be necessary to cover a large part of the accessible phase space, so either trajectories must be very long or we must use many initial conditions. There are many ways to solve initial value problems such as (2.1) combined with an initial condition $z(0) = \boldsymbol{\zeta}$. The methods introduced here all rely on the idea of a discretization with a finite stepsize $h$, and an iterative procedure that computes, starting from $z_0 = \boldsymbol{\zeta}$, a sequence $z_1, z_2, \ldots$, where $z_n \approx z(nh)$. The simplest scheme is certainly Euler's method which advances the solution from timestep to timestep by the formula[1]

$$z_{n+1} = z_n + h\boldsymbol{f}(z_n).$$

The method is based on the observation that $z(t+h) \approx z(t)+h\dot{z}(t)$, i.e. the beginning of a Taylor series expansion in powers of $h$, and the further observation that the solution satisfies the differential equation, hence $\dot{z}(t)$ may be replaced by $\boldsymbol{f}(z(t))$.

In order to be able to easily compare the properties of different methods in a unified way, we focus in this chapter primarily on a particular class of schemes, generalized *one-step methods*. Suppose that the system under study has a well defined flow map $\mathscr{F}_t$ defined on the phase space (which is assumed to exclude any singular points of the potential energy function). The solution of the initial value problem, $\dot{z} = \boldsymbol{f}(z)$, $z(0) = \boldsymbol{\zeta}$, may be written $z(t, \boldsymbol{\zeta})$ (with $z(0, \boldsymbol{\zeta}) = \boldsymbol{\zeta}$), and the flow-map $\mathscr{F}_t$ satisfies $\mathscr{F}_t(\boldsymbol{\zeta}) = z(t, \boldsymbol{\zeta})$. A *one-step method*, starting from a given point, approximates a point on the solution trajectory at a given time $h$ units later. Such a method defines a map $\mathscr{G}_h$ of the phase space as illustrated in Fig. 2.1.

We assume here a basic understanding of ordinary differential equations; some good references for review of this topic are [16, 51, 177, 362]. For basic concepts in the numerical analysis of ordinary differential equations the definitive reference is [167].

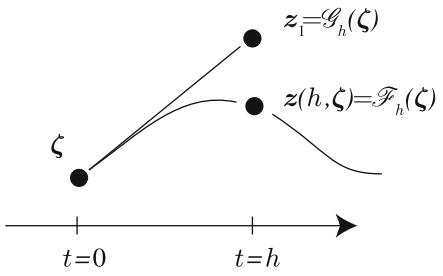



**Fig. 2.1** A step with the flow map approximation $\mathscr{G}_h$ is illustrated in comparison to the corresponding step along the solution curve defined by the flow map $\mathscr{F}_h$

[1] Subscripts were used previously to indicate the components of vectors and here they are used to indicate the indexing of timesteps. Although in theory this could lead to some confusion, it normally does not in practice, since we index components in descriptions of details of models and we discuss timesteps in context of defining numerical methods for general classes of systems. Moreover, we use boldface for vectors, so a subscript on a boldface vector indicates a timestep index. When we wish to refer to both the timestep and the component index, we may write $z_{n,i}$ to denote the $i$th component at timestep $n$.